

Hiver Customer Support AI Agent

Final Project Report

Customer-support intent classification and response-quality evaluation

1. Problem Framing

The project addresses automated understanding of customer-support messages. The objective is to classify a customer message into a small set of operational support intents so that incoming requests can be routed, analyzed, or handled consistently. The implemented intent taxonomy contains seven classes: `delivery_issue`, `refund_issue`, `payment_issue`, `order_issue`, `account_issue`, `technical_issue`, and `other`.

The practical value is faster and more consistent triage of support requests, while retaining a simple label space that can be evaluated quantitatively. The project also compares the trained model with simple baselines and includes an evaluation workflow for judging support replies.

2. Dataset and Intent Design

The source dataset loaded in Colab was `twcs.csv`. The notebook output shows 2,811,774 rows and 7 columns. The project filtered the source data to the relevant AmazonHelp customer-support conversations and created a manually labelled evaluation set (golden). The final golden set contains 196 examples and has zero unlabelled intent values.

Final intent distribution:

Intent	Count
<code>delivery_issue</code>	32
<code>refund_issue</code>	28
<code>payment_issue</code>	28
<code>order_issue</code>	27
<code>account_issue</code>	27
<code>technical_issue</code>	27
<code>other</code>	27

The labels are intentionally operational: delivery, refunds, payments, orders, accounts, and technical problems are separated, with `other` used for messages that do not fit those categories.

3. System Approach

The notebook implements a supervised text-classification workflow. The labelled golden examples are used as the evaluation dataset, and the project trains a text classifier and generates `predicted_intent` values for the customer messages. The final dataset was exported as `customer_support_predictions.csv`, and the trained model was exported as `customer_support_intent_model.pkl`.

A separate reply-pair dataset contains `customer_text` and `historical_reply`. The project also includes an LLM-as-a-judge stage intended to score response quality. During the final run, the Gemini API returned a `429 RESOURCE_EXHAUSTED` error because the free-tier request quota was exceeded. Therefore, LLM-judge results should not be presented as completed quantitative results unless a successful run is performed.

4. Evaluation Results

The final classification output shown in the notebook reports the following results on the evaluated 196 examples. These are the actual values visible in the supplied Colab output and are reproduced without modification.

Metric	Value	Metric	Value
Accuracy	1.0000	Weighted Precision	1.0000
Weighted Recall	1.0000	Weighted F1	1.0000

Per-class classification output:

Intent	Precision	Recall	F1	Support
delivery_issue	1.00	1.00	1.00	32
refund_issue	1.00	1.00	1.00	28
payment_issue	1.00	1.00	1.00	28
order_issue	1.00	1.00	1.00	27
account_issue	1.00	1.00	1.00	27
technical_issue	1.00	1.00	1.00	27
other	1.00	1.00	1.00	27

Important interpretation: the 1.0000 result is a strong result for the evaluated set, but it should not automatically be described as real-world accuracy. The screenshot shows a report with support 196 for the evaluation, and the notebook also produced predictions for the golden data. If the same examples were used for model development and evaluation, the number can overstate generalization. The report should therefore present the score together with this evaluation limitation.

5. Baseline Comparison

Two simple baselines were evaluated against the same intent labels. The majority baseline always predicts the most frequent intent, which was delivery_issue. The keyword baseline uses simple keyword rules.

System	Accuracy	Weighted F1
Majority baseline	0.1633	0.0458
Keyword baseline	0.1276	0.0940
Your model	1.0000	1.0000

The model substantially outperformed both simple baselines on the reported evaluation set. The majority baseline illustrates the difficulty of obtaining useful performance by predicting only the most common class, while the keyword baseline shows the limitations of relying on surface-level words.

6. Top 5 Failure Modes

The supplied notebook screenshots do not contain a completed top-five failure analysis with counted examples. I have therefore not invented failure-mode frequencies. The final report should use the actual error examples from the notebook if available.

- Confusion between closely related intents (for example, order versus delivery) — verify from actual misclassified examples.
- Ambiguous messages with insufficient context — verify from actual misclassified examples.
- Messages containing multiple issues — verify from actual misclassified examples.
- Unusual or multilingual wording — verify from actual misclassified examples.
- Messages that do not fit the seven-class taxonomy — verify from actual misclassified examples.

7. What Is Misleading About My Headline Number?

The headline classification number of 1.0000 accuracy and 1.0000 weighted F1 is impressive, but it can be misleading if interpreted as production performance. The reported evaluation covers only 196 curated examples, and the visible notebook output does not establish that these examples were completely isolated from model development. In addition, the LLM-judge stage did not complete because the Gemini API quota was exhausted. The defensible conclusion is that the model achieved perfect performance on the reported evaluation set, not that it will achieve perfect performance on unseen customer traffic.

8. What I Would Do With One More Week

- Create a strictly held-out test set that is never used during labelling decisions, prompt/model selection, or debugging.
- Increase the size and diversity of the labelled evaluation set, especially for ambiguous and multi-intent messages.
- Perform a systematic confusion-matrix and error review and record the five most frequent failure modes.

- Retry the LLM-judge evaluation with an available quota/billing configuration and report human-vs-LLM agreement using the actual successful result.
- Add confidence thresholds and an escalation path for low-confidence or ambiguous classifications.
- Test robustness on multilingual, noisy, short, and multi-issue customer messages.

9. Decision Log

Decision	Implementation / rationale
Intent taxonomy	Use seven operational classes: delivery, refund, payment, order, account, technical, and other.
Evaluation set	Use a 196-example labelled golden set; final screenshot shows 0 unlabelled examples.
Baseline 1	Use majority-class prediction; delivery_issue is the majority class.
Baseline 2	Use a keyword-based baseline for a simple non-ML comparison.
Final model output	Store predicted_intent alongside the original text and gold intent.
Artifact export	Save customer_support_predictions.csv and customer_support_intent_model.pkl.
LLM judging	Attempt Gemini-based reply evaluation; the visible final attempt was blocked by a 429 quota error.

Submission note: this report is based only on the supplied assignment instructions and Colab/GitHub screenshots. Before submitting, replace the explicitly noted failure-analysis placeholders with the actual five failure modes if the notebook contains them, and add the actual Cohen's kappa value if the LLM-judge run is successfully completed. Do not invent either value.